MINIT Ignífugos

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COPIA FIEL DE CERTIFICADOS **HIDRO-LACA MINIT**

Presentación valida únicamente con el Certificado / Factura de Compra

validacion

LISTADO DE ENSAYOS APROBADOS

INTI OT: 101-26333

IRAM 11910-3 / NBR 9442/86 / ASTM 162. Inflamabilidad de materiales mediante el método de panel radiante que expone una muestra de tamaño reducido a una fuente de calor radiante y un quemador piloto, midiendo la propagación de la llama y la evolución del calor para obtener un índice de inflamabilidad. El objetivo es determinar la capacidad de un material para mantener una llama y el calor que genera en condiciones de laboratorio.

CONICET - INFORME ScienceDirect Construction and Building Materials 301 (2021) 124058

- **UNE-EN 13501-1:** establece la clasificación de la **reacción al fuego de productos de construcción y elementos para edificación** , utilizando un sistema de "Euroclases" (de A a F) que evalúa la inflamabilidad, la producción de humo y las gotas incandescentes.. Es la normativa europea vigente que se aplica en España para evaluar la seguridad contra incendios de materiales como tejidos ignífugos, barnices y otros productos utilizados en construcciones.

- **IEC 60695-2-11:** es norma internacional que especifica el método de ensayo de inflamabilidad con hilo incandescente para productos finales, subconjuntos y componentes de equipos electrotécnicos. Simula un riesgo de incendio aplicando un hilo incandescente caliente a una muestra para comprobar si se autoextingue o si, tras la ignición, su capacidad de propagación de la llama es limitada. La norma se utiliza para evaluar el riesgo de incendio midiendo la ignición y la propagación de la llama.

- **ISO 844:** Se describe como Método B, medición directa de la deformación de compresión en la sonda. Este método B aporta datos sobre la deformación y un módulo de compresión menos influenciado por los efectos de aplicación de fuerza.

- **ISO 4898:** Establece los requisitos y métodos de ensayo para productos de aislamiento térmico de **Plásticos celulares rígidos** utilizados en la construcción. Su objetivo es especificar las propiedades físicas de estos materiales, como la conductividad térmica, densidad y resistencia a la compresión, para garantizar su uso en aplicaciones de aislamiento.

- **ISO 11925-2**: documenta la prueba de inflamabilidad de materiales bajo la exposición directa de una llama, que evalúa la facilidad con que un producto se inflama.

- **ISO 8990**: determina las propiedades de transmisión térmica en régimen estacionario utilizando el método de la **caja caliente calibrada y guardada**. Este tipo de informe, analiza el flujo de calor a través de un material para calcular su resistencia y transmitancia térmica.

Marco de Reconocimiento Internacional (Reciprocidad)

El sistema de aceptación de ensayos extranjeros en Argentina se basa en acuerdos internacionales de reciprocidad que eliminan la necesidad de repetir pruebas locales. Recientemente, el marco normativo se ha simplificado mediante el Decreto 892/2025.

El Gobierno argentino estableció el reconocimiento automático de certificaciones y ensayos internacionales para agilizar el comercio:

- Se consideran válidos los ensayos realizados en países con altos estándares de vigilancia técnica (países de "alta vigilancia") detallados en la normativa.

- Los productos que posean certificaciones acreditadas por entidades avaladas por acuerdos de reciprocidad no requieren nuevos ensayos locales.

- Rol del INTI y la OAA: Mientras el INTI se enfoca en la metrología científica y legal, el OAA es el encargado de mantener los vínculos de reciprocidad con los organismos de acreditación extranjeros.

Principales Organismos de Acreditación y Cooperación Regional e Internacional

Organismo	Alcance	Rol Principal
ILAC	Global	Cooperación para laboratorios de ensayo y calibración.
IAF	Global	Reconocimiento de organismos de certificación de productos y sistemas.
IAAC	Regional (Américas)	Cooperación Interamericana de Acreditación.
OCDE	Global	Acuerdos de "Aceptación Mutua de Datos" (MAD) para seguridad ambiental y sanitaria.
A2LA	Regional (Américas)	Asociación para la Acreditación de Laboratorios. Bajo la norma ISO/IEC 17025. Garantiza la competencia técnica y la confianza global
ANAB / ANSI	Regional (Américas)	Junta de Acreditación. Organismo de acreditación de normas, laboratorios y organismos de inspección y certificación (ISO/IEC 17025, 17020, 17021, 17034).

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IRAM 11910-3 NBR 9442/86 ASTM 162 (METODO PANEL RADIANTE EN MADERA, RE3 CLASE B, BAJA PROPAGACION DEL FUEGO)

Para tratamiento por recubrimiento de superficie con lacas

Informe de Ensayo

OT N°101 - 26333 Único
Página 1 de2

Fecha de Informe: 15/03/2016

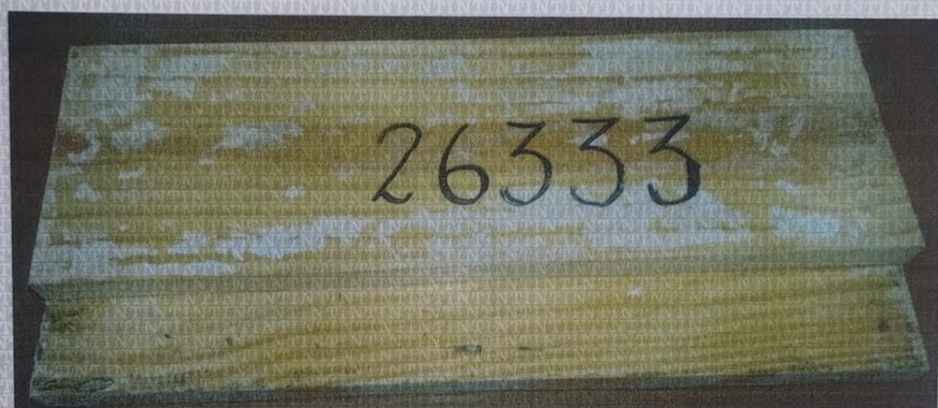
Solicitante

Minotti Pablo David

Echeverria 2073 - Piso 4 Dpto. B - (1419) – Ciudad Autónoma de Buenos Aires.

Elemento

Una (1) muestra de madera con tratamiento, identificada por el solicitante como: **"Tablas de madera tratadas con Aditivo Primer Intumescente MINIT para lacas y pinturas al agua y lacas".**



Determinaciones requeridas

Clasificación de acuerdo al Índice de Propagación de Llama.

Fecha de Recepción

03/02/2016

Fecha de ensayo

10/03/2016

Metodología empleada

El ensayo de Propagación Superficial de Llama se realizó de acuerdo a la **Norma IRAM 11910-3:1994 "Materiales de Construcción, Reacción al fuego, Determinación del índice de propagación de llama – método del panel radiante"** (coincide con los métodos de ensayo de la Norma NBR 9442:1986 y ASTM E162:1994).

INTI Construcciones

Presidencia de la Nación | Ministerio de Producción

Informe de Ensayo

OT N°101 - 26333 Único
Página 2 de2

Resultados

F(promedio):	3,09
Q(promedio):	9,57
I(promedio):	**30,23**

Teniendo en cuenta la Tabla de Clasificación de la Norma IRAM 11910-1 del año 1994, el **Índice de Propagación de Llamas (I)** hallado del material denominado: **"Tablas de madera tratadas con Aditivo Primer Intumescente MINIT para lacas y pinturas al agua y lacas"** se clasifica como:

"Clase RE 3: Material de Baja propagación de llama"
(A esta clase pertenecen los materiales con un índice entre 26 y 75)
Coincide con la Clase B de la Norma brasileña NBR 9442/1986

Referencias para el ensayo de determinación de la propagación superficial de llama

Clase	Clase ABNT	Denominación	Norma IRAM	Criterio de clasificación
RE 1	-	Incombustible	11910-2	Anexo A de la norma
RE 2	A	Muy baja propagación de llama	11910-1	Índice: 0 a 25
RE 3	B	Baja propagación de llama	11910-1	Índice: 26 a 75
RE 4	C	Mediana propagación de llama	11910-1	Índice: 76 a 150
RE 5	D	Elevada propagación de llama	11910-1	Índice: 151 a 400
RE 6	E	Muy elevada propagación de llama	11910-1	Índice mayor a 400

Definiciones:
Un factor derivado de la rapidez de propagación del frente de llama (F) y otro relativo al calor liberado por el material ensayado (Q) son combinados para proveer el Índice de propagación superficial de llama (I).
I: Índice de propagación superficial de llama.
F: Factor de propagación de llama.
Q: Factor de evolución de calor

Los resultados contenidos en el presente informe corresponden a las condiciones en las que se realizaron las mediciones y/o ensayos.

Fin del Informe

Arq. BASILIO HASAPOV
COORDINADOR
U.T. TECNOLOGIA EN INCENDIOS
INTI-CONSTRUCCIONES

Arq. INÉS DOLMANN
DIRECTORA TECNICA
INTI - Construcciones

IRAM 11910-3 / NBR 9442/86 ASTM 162 Tabla de Resistencia OT Nº: 101-26333

La norma ASTM E162 evalúa la inflamabilidad de los materiales mediante un método de panel radiante que expone una muestra a una fuente de calor radiante y un quemador piloto, midiendo la **propagación de la llama y la evolución del calor** para obtener un índice de inflamabilidad. Determina la capacidad de un material para mantener una llama y el calor que genera en condiciones de laboratorio.

En qué consiste la prueba ASTM E162:

1. **Objetivo:** Evaluar y medir la inflamabilidad de la superficie de los materiales, particularmente para productos de construcción y en la industria del transporte.

2. **Equipo:** Se utiliza un equipo de panel radiante que incluye:

- Un panel radiante de gas como fuente de calor a una temperatura a 1238 ºF.
- Un quemador piloto que enciende la llama en la muestra.
- Un soporte porta-muestras para colocar la muestra de 6 x 18 pulgadas (aprox. 15 x 46 cm).
- Un sistema de extracción de humos y un puesto de control con software para el registro de datos.

3. **Procedimiento:**

- Se coloca una muestra del material de prueba en el soporte.
- Se expone la muestra a la fuente de calor radiante durante un máximo de 15 minutos.
- Se registra el progreso de la propagación de la llama y la evolución del calor que emite la muestra.

4. **Resultados:** La prueba genera un índice de inflamabilidad que se calcula a partir de la **propagación de la llama y la cantidad de calor** que la muestra ha liberado.

En base al resultado obtenido en este ensayo: Case A OT: 101-26332 con dos capas de 80 micrones, expuesta por 15 minutos de revestimiento (pintura) ignifugo intumescente aislante térmico MINIT:

Clasificaciones: Clase B / RE3 **Q** (Factor de evolución de calor) **9.57** × **F** (Factor de propagación de la llama) **3.09**
¨I¨ Promedio = 30.23

Calculo de Desempeño en base a resultado IRAM 11910

Sistema	Nivel	Espesor seco (µm)	Manos	Consumo (kg/m²)	Desempeño estimado	Clasificación referencial
MINIT Laca	Protección inicial	80 µm	2 manos	0.08	Retardo de llama ~15 min	RE 3
MINIT Laca	Protección reforzada	120–150 µm	3 manos	0.12–0.15	Retardo medio ~25–30 min	RE 2 mejorado
MINIT Laca	Protección avanzada	180–220 µm	4–5 manos	0.18–0.22	Retardo alto ~30–45 min	RE 2 (optimizado)

Informe del CONICET y otros, Publicado en Revista *Science Direct*

Traducción Resumida

Instituto de Investigaciones y Políticas del Ambiente Construido, Facultad de Arquitectura y Urbanismo, Universidad Nacional de La Plata, Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET) 1900 La Plata, Argentina.

*** Materiales de Construcción y Edificación**

Los ensayos realizados con **Aditivo para pinturas retardante de fuego, de base acuosa (MINIT Ignífugo de Pablo Minotti)** sobre materiales combustibles cumplen con las siguientes normas internacionales:
UNE-EN 13501-1 / IEC 60695-2-11 / ISO 844 / ISO 4898 / ISO 11925-2 / ISO 8990

La prueba de flamabilidad demostró que los materiales de base a proteger pueden resistir sobrecalentamiento de 960℃

2.3. Características de la flamabilidad y la combustibilidad
Se llevaron a cabo dos pruebas para evaluar la reacción en los materiales de aislamiento frente a diferentes situaciones de riesgo. Se utilizó la prueba de flamabilidad, la cual simula los efectos de una ***tensión térmica*** para reproducir el riesgo de un incendio a través de un hilo incandescente. Al mismo tiempo, la prueba de combustibilidad se llevó a cabo para medir la expansión de una llama, que simula el comienzo de un incendio. Se prepararon dos muestras para cada prueba respectivamente: 150 mm 150 mm 100 mm y 90 mm 250 mm 40 mm. Las pruebas evaluaron los materiales mencionados en la sección 2.1 (muestras de tipo 1), más ocho muestras con diferentes aditivos y revestimientos de pintura detallados en la fig.3. Para las muestras de tipo 2, se utilizó un **aditivo para pinturas retardante de fuego, (MINIT Ignífugo de Pablo Minotti)**. Las muestras de tipo 3 a 7 utilizaron esmalte sintético estándar (compuesto por 66,2% de resina alquídica, 23% de solvente Stoddard, pigmentos y rellenos inertes) y látex (compuesto por 68,5% de polímero acrílico, 13% de agua, 12% carbonato de calcio, pigmentos, rellenos inertes, aditivos y agentes coalescentes) mezclado con solubles como arena, bórax o **aditivo retardante de fuego, para pinturas (MINIT Ignífugo de Pablo Minotti)**. Cabe aclarar que algunos de estos materiales habían sido utilizados como retardantes de fuego en estudios previos [45]. Por último, las muestras de tipo 8 fueron revestidas con látex y las tipo 9 con cemento preparado con componentes y proporciones de la mezcla. **Todos los revestimientos se aplicaron manualmente con un pincel en una sola capa, con la opción de utilizar pintura en spray para la aplicación del látex en las tipo 8**.

3.1. Propiedades aislantes térmicas
Se midió la conductividad térmica en tres muestras de materiales de base según la sección 2.1. Cada muestra se analizó en un recipiente caliente en base al **estándar de ISO 8990** [44]. La conductividad térmica tuvo un rango de 0,0603 y 0,0706 W/m K. Estos valores fueron calculados con las ecuaciones 1 y 2 con las medidas recolectadas. Estas se pueden observar en la tabla 2. Los materiales tuvieron una densidad promedio de 168 ± 7 kg/m3. La variedad de los valores entre la conductividad térmica y la densidad se puede asociar con los métodos manuales utilizados para analizar las muestras. Puede haber variaciones en la cantidad de material que se utilizó en cada molde, lo que puede modificar la compresión, la proporción de aire en el molde y la capacidad de aislamiento térmico en cada muestra. Se comparó la conductividad térmica y la densidad promedio con los materiales convencionales y alternativos que actuaron como aislantes (Fig.8). Estos valores de conductividad son similares pero más altos a aquellos obtenidos con los materiales de base. Esto sugiere que este material presenta una conductividad apropiada para el uso de aislamiento térmico.

3.2. Flamabilidad y combustibilidad
Los materiales mencionados en la sección 2.1 y las ocho muestras con **aditivo retardante de fuego, para pinturas (MINIT Ignífugo de Pablo Minotti)** y revestimientos detallados en la sección 2.3 fueron evaluados en base a su flamabilidad y su combustibilidad. **Los materiales se evaluaron con un hilo incandescente aplicado a 840℃ y 960℃, acorde al estándar internacional IEC 60695-2-11** [46]. Los nueve tipos de muestras pasaron la aplicación
a 840℃ y 960℃. La fig. 9 muestra la respuesta de las muestras durante la aplicación en ambas temperaturas. A los 840℃, **las muestras pasaron la prueba, ya que el hilo no incendió ni el material ni el filtro de papel debajo de la muestra**. A su vez, tanto el tipo 1 como las **muestras revestidas con látex y aditivo retardante de fuego, para pinturas (MINIT Ignífugo de Pablo Minotti)** (tipo 5), solo látex (tipo 8) y solo cemento (tipo 9) **no se incendiaron ni se les desprendieron partículas durante la aplicación a los 960℃**. La tabla 3 representa las muestras que sí se incendiaron. Podemos resaltar que en estas muestras, la combustibilidad fue extinguida rápidamente y no hubo partículas que encendieran el filtro de papel. Por lo tanto, incluso este tipo de muestras pasaron la prueba de los 960℃. A continuación, se siguieron evaluando todos los tipos de muestras con la **prueba de la combustibilidad de ISO 11925-2** [47]. Las muestras de tipo 1 se prendieron fuego, sin embargo, la llama tuvo poco alcance gracias al cemento que actuó como retardante. El efecto del cemento mezclado con EPS también fue observado en otras investigaciones [40]. Al mismo tiempo, se observó que mientras los gránulos de EPS se consumieron, la cobertura no produjo cenizas. Además, la propagación de la llama no excedió los 150 mm de altura durante los 30 segundos en los que se aplicó el fuego. Por lo tanto, el material obtendría por lo menos la clasificación E en base al **estándar de la UNE-EN 13501-1** [48]. Los materiales clasificados como E pueden resistir una llama pequeña durante un periodo corto de tiempo sin que la llama se esparza substancialmente. Sin embargo, la única excepción son los tipos 3 y 4 con revestimientos de esmalte sintético con bórax y arena respectivamente, debido a que la llama excedió el máximo permitido. La fig. 10 muestra las llamas con la altura máxima y cómo quedó el área afectada, la cual se observó en las tres repeticiones llevadas a cabo para cada variante. **Las muestras que no fueron afectadas y que tuvieron la menor cantidad de partículas desprendidas fueron las de látex y aditivo retardante de fuego, para pinturas (MINIT Ignífugo de Pablo Minotti)** (tipo 5). **Por lo tanto, este revestimiento mejora el rendimiento de los materiales de base**. **Se considera apropiado aplicar este revestimiento cuando se requiere protección adicional contra el fuego, debido a las condiciones en las que se aplica el material de aislamiento.**

Ensayos Aprobados:

UNE-EN 13501-1: establece la clasificación de la **reacción al fuego de productos de construcción y elementos para edificación** , utilizando un sistema de "Euroclases" (de A a F) que evalúa la inflamabilidad, la producción de humo y las gotas incandescentes.. Es la normativa europea vigente que se aplica en España para evaluar la seguridad contra incendios de materiales como tejidos ignífugos, barnices y otros productos utilizados en construcciones.

IEC 60695-2-11: es norma internacional que especifica el método de ensayo de inflamabilidad con hilo incandescente para productos finales, subconjuntos y componentes de equipos electrotécnicos. Simula un riesgo de incendio aplicando un hilo incandescente caliente a una muestra para comprobar si se autoextingue o si, tras la ignición, su capacidad de propagación de la llama es limitada. La norma se utiliza para evaluar el riesgo de incendio midiendo la ignición y la propagación de la llama.

ISO 844: Se describe como Método B, medición directa de la deformación de compresión en la sonda. Este método B aporta datos sobre la deformación y un módulo de compresión menos influenciado por los efectos de aplicación de fuerza.

ISO 4898: Establece los requisitos y métodos de ensayo para productos de aislamiento térmico de **Plásticos celulares rígidos** utilizados en la construcción. Su objetivo es especificar las propiedades físicas de estos materiales, como la conductividad térmica, densidad y resistencia a la compresión, para garantizar su uso en aplicaciones de aislamiento.

ISO 11925-2: documenta la prueba de inflamabilidad de materiales bajo la exposición directa de una llama, que evalúa la facilidad con que un producto se inflama.

ISO 8990: determina las propiedades de transmisión térmica en régimen estacionario utilizando el método de la **caja caliente calibrada y guardada**. Este tipo de informe, analiza el flujo de calor a través de un material para calcular su resistencia y transmitancia térmica.

4. Conclusiones
Los resultados obtenidos en las pruebas de caracterización y sus variantes con los revestimientos de pintura y el **aditivo retardante de fuego, para pinturas (MINIT Ignífugo de Pablo Minotti)** nos permitieron llegar a las siguientes conclusiones. La prueba de flamabilidad demostró que el material de base y sus variantes pueden resistir el sobrecalentamiento hasta los 960℃. La prueba de combustibilidad demostró que los materiales de base no tienen riesgo de la expansión de una llama única. Por lo tanto, se considera que por lo menos tienen una clasificación E de acuerdo al estándar de la UNE-EN 13501-1. En caso de que se requiera reforzar el material, el rendimiento puede ser mejorado con la aplicación de ciertos revestimientos, tales como el **látex** y **aditivo retardante de fuego (MINIT Ignífugo de Pablo Minotti) con su óptima respuesta contra el fuego**. El rendimiento del aislamiento del material sugiere que este es factible para varios tipos de aplicaciones.

A continuación publicación científica completa.

Construction and Building Materials 301 (2021) 124058

Contents lists available at ScienceDirect

Construction and Building Materials

journal homepage: www.elsevier.com/locate/conbuildmat

Characterization of an alternative thermal insulation material using recycled expanded polystyrene

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highlights

- Easy-to-replicate recycled EPS based thermal insulation material.
- The thermal insulation material is effective and capable of resisting fire risks.
- The mechanical properties make it suitable for non-load bearing applications.
- Incorporation of coatings and additives improves the performance of the material.
- Safe thermal insulation applications are envisioned for the low-income population.

article info

Article history:
Received 30 December 2020
Received in revised form 10 June 2021
Accepted 20 June 2021

Keywords:
Recycled EPS insulation
Portland cement
Thermal conductivity
Low-cost insulation material

abstract

Thermal insulations help to reduce household energy demand and extend the periods of thermal comfort generated by heating and cooling systems. Conventional isolations may not be economically accessible to all social sectors, despite their benefits. This paper presents and evaluates an alternative low-cost insu-lating material and its variants, based on the mechanical reuse of expanded polystyrene (EPS) waste packaging, cementitious binder, plastic additives, and water. These materials show comparable perfor-mance to commercially available insulations (thermal conductivity range 0.0603–0.0706 W/m K) and provide an environmentally-safe and cost effective alternatives for house isolation for the low income population living in settlements worldwide.

1. Introduction

Many citizens do not access to adequate housing and basic services and live in informal conditions. United Nations (UN-Habitat) defines informal settlements as urbanizations that lack one or more of the following conditions or services: formal land tenure in suitable areas for housing; adequate living space; safe and permanent housing; tap water and sewer services. According to UN-Habitat, 881 million urban citizens lived in informal urban-izations in developing countries during 2014. Just over 100 million people, around 20% of the total population lived in this informal condition in Latin America and the Caribbean [1]. In Argentina where this research takes place, the National Registry of Popular Settlements (RENABAP) refers to popular settlements as those made up of at least 8 families where more than half of the popula-tion has no property title to the land and lacks two or more of the basic services. In this terms Argentina has around 4,400 popular settlements where>900,000 families reside, as stated RENABAP in the latest report prepared in 2020 [2]. Our research focuses on the district of La Plata, which registers the largest number of infor-mal urbanizations in the province of Buenos Aires. In 2018 RENA-BAP counted in this location, a total of 153 popular settlements hosting around 29,000 families [3].

Housing solutions with very low investment predominate in this context. These are based on self-construction without profes-sional assistance and usually have low efficiency and durability. The National Institute of Statistics and Censuses of the Argentine Republic (INDEC) registered in 2010 that, according to the quality of the materials, 62.6% of the total houses of the country had an acceptable quality, 33.3% were in recoverable condition and the 4.1% were in unrecoverable condition [4]. Currently these condi-tions have been further aggravated by the covid-19 pandemic.

The use of enclosures with insufficient or no thermal insulation is one of the construction problems in popular settlements. Metal veneer and wood are the most frequently used materials, and

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https://doi.org/10.1016/j.conbuildmat.2021.124058 0950-0618/

bricks without plaster in some better cases. This condition affects the habitability and thermal comfort, as well as the rational use of energy [5,6]. Therefore, it is a problem that justifies the research of conventional and alternative (naturally or recycled) insulation materials to propose solutions for the most vulnerable social groups.

Insulation materials are characterized for presenting low ther-mal conductivity. This parameter expresses the ability of materials to transfer heat flows. Mineral wools and plastic foams have a good thermal performance in these terms [6–10] and are the most widely used conventional insulating materials around the world [11]. Despite their high efficiency, they produce unfavorable envi-ronmental impacts during their manufacture and contribute to the generation of non-degradable waste [12]. On the contrary, materi-als of natural origin are an environmentally friendly alternative source for building insulations. Crop harvests residues or by-products stand out in this category [12–16]. As a disadvantage, these materials have a limited ability to be scaled up for serial pro-duction, due to their temporary availability and the required selec-tion and conditioning for an optimal performance. Insulation materials generated from recycling of high-demand industrial products are more readily available than natural materials and have low or marginal cost. In addition, recycling generates envi-ronmental benefits by reducing the need to produce new raw materials, harnessing the potential from discards that could other-wise end up in landfills [17–24]. It is considered that the use of these recycling materials would be an accessible and beneficial option both for the improvement of housing deficiencies of most vulnerable social sectors, as well as for generation of job opportu-nities by adding value to waste materials. Hence, our group ana-lyzed alternative low cost and widely available insulation materials, which were feasible to be used by social sectors with fewer economic resources. Materials grouped in plastics, natural fibers, papers, and soils were initially evaluated [25]. Their thermal conductivity was quantified using an equipment developed for its analysis. Subsequently, the study focused on the use of expanded polystyrene (EPS) [26].

EPS is characterized by a cellular structure composed mostly of hermetic air (around 96–98%) that causes its good thermal insula-tion capacity with a commercial thermal conductivity between 0.03 and 0.04 W/m K measured in previous research [27–28]. This characteristic hints that EPS from packaging is one of the recycled materials that can be used as thermal insulation while benefiting the environment. Currently, packaging industrial sector provides the largest amount of plastic waste worldwide, representing 46% of the total industrial sector, producing 141 million tons of waste annually. These products have the shortest service life, averaging only 0.5 years [29]. Some studies have proposed the incorporation of crushed or mechanically recycled EPS for housing applications. This material is most commonly used as a component in the prepa-ration of lightweight concrete [30–37]. Load-bearing panels or enclosure panels have been developed and evaluated with this type of mixture. These applications with low proportion of EPS, have good mechanical resistance but they show lower insulating capacity than commercial insulations. Other studies proposed Portland cement with natural resin [35] or plaster [38] as EPS bin-ders. The authors concluded that as the proportion of EPS in the mixture increases, thermal conductivity, density and mechanical strength decrease. There are fewer studies of materials made with large amounts of EPS and little binder [38–40]. These materials were suggested for the manufacture of mortars, insulation filler or panels, but they have not led to specific applications.

In Argentina, EPS waste does not have a specific post-consumer destination and it is widely available as an urban waste. Due to this fact, our group has investigated the use of recycled and commercial EPS for the elaboration of an alternative insulation using a binder mixture based on Portland cement [26]. We proposed and charac-terized a variant with optimal insulating performance, which is composed of fine and coarse granules of mechanically recycled EPS. The evaluation of this material is necessary to advance in the development of a suitable component for homes. This is impor-tant considering that houses in popular settlements have problems of water leaks as well as fires due to the low condition of the elec-trical systems. The insulation must also be easy to adapt for its placement and application in different types of housing.

The goal of this paper is to evaluate the performance of an alter-native insulating material generated from recycled EPS bound with a cement-based mix, and its different variants with coatings and additives. Their performance is evaluated in terms of thermal insu-lation capacity, flammability and ignitability, mechanical proper-ties, water absorption, weathering behavior and machinability. The purpose of this study is to contribute to the development and characterization of a component that improves the habitability of homes in popular sectors that are in recoverable conditions or houses that are going to be built, by adding a reduced investment. Hence for the elaboration of this material, accessible components, tools and simple production methods are proposed, so it can be self-constructed by the social sectors with fewer resources.

2. Materials and methods

2.1. Composing and molding the alternative insulation material

Recycled EPS containers were the raw material used for the pro-posed alternative insulation. EPS waste was recycled mechanically using a low-cost and easy-to-build crusher model [41,42] that replaced the previous manual method [26]. This crusher has a rotating blade and a receptacle in its upper part through which the EPS waste is supplied. A metal mesh defined the size of the resulting granules with an average density of 18 ± 1 kg / m^3. The size and gradation of the EPS granules was measured using differ-ent sieves (Fig. 1).

The alternative material was formulated seeking to increase its insulation capacity. Previous research has shown that thermal con-ductivity depends on the density of the material [39]. This led us to test a mix with maximal proportions of recycled EPS and minimal proportions of the cementitious binder. The proportions of the bin-der were set in such a way that the mixture was neither light nor thick, to facilitate its wet integration with the EPS granules and based on the cohesion of the final dry material. The volume ratio in the binder mixture of Portland cement, water and multipurpose vinyl additive was 4:3:1/3. This vinyl additive is composed of 10–40% polyvinyl acetate-based copolymer, water, and organic and inorganic additives (Tacuru Weber Saint-Gobain) [43]. The additive was incorporated to improve the adhesion of the EPS based on the amounts recommended by the manufacturer. The dose of binder in relation to the EPS was determined based on the texture of the dry material on the surface opposite to the pour-ing side. The mix was considered adequate when crushed EPS granules could be seen on this surface and no excess binder film was visible on the final material. Table 1 shows the quantities and the volume percentages for each component per m^3 to achieve a material with appropriate characteristics. The binder compo-nents were blended until a homogeneous mixture was obtained. EPS particles were then added, which constituted about 90% of the total volume and did not float in the mix. These components were mixed until homogeneous before pouring the mixture into moulds. After 72 h in an indoor environment at room temperature, the moulds were removed and the samples were kept in this con-dition for approximately 5 more days for final drying. For some tests, variants of this base material were also carried out with addi-

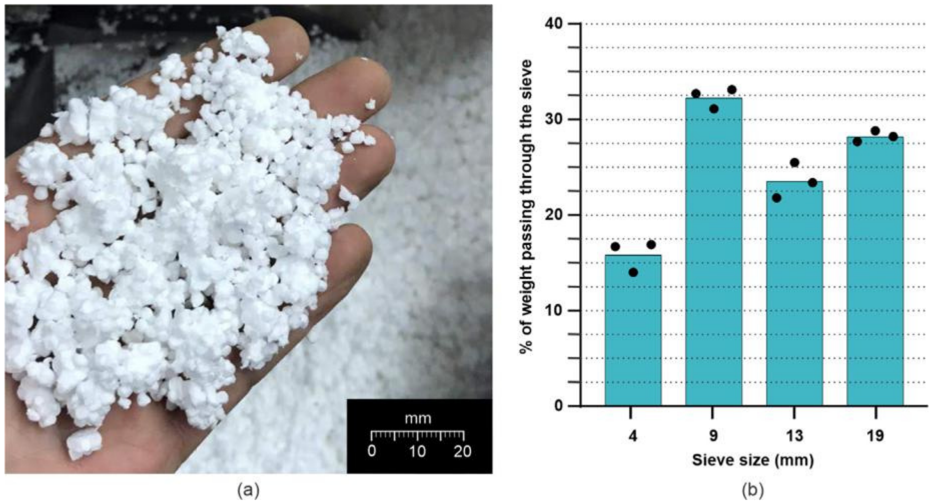

Fig. 1. EPS granules resulting from mechanical recycling (a) and its gradation (b).

Table 1
Materials required per m^3 to produce thermal insulation samples.

Material	Quantity per m^3	% by volume
Recycled EPS	17.46 kg	90.63
Portland cement	142.00 kg	5.11
Water	76.00 l	3.83
Vinyl additive	8.46 l	0.43

tives in the binder mixture or with coatings once the material had dried.

2.2. Characterization of insulation capacity

Thermal conductivity and thermal transmittance were mea-sured using a calibrated hot box testing apparatus developed in previous research [25] based on ISO 8990 standard [44]. A hot box with dimensions of 820 mm 540 mm 830 mm, was used to measure the steady state heat flow passing through a test sam-ple placed between a hot and a cold source using the heat flow meter (HFM) method. The cold source was a commercial refriger-ation equipment (Hma 122Fc B Mabe). The hot source was an elec-trical resistance that comes from a commercial heating pad (Large AL81 SILFAB) with a copper plate of 400 mm 400 mm to generate even heat distribution. Both the hot source and the sample were insulated with commercial EPS.

Thermal conductivity (W/m K) was measured in three samples that weremanufactured ina dimensionof 400 mm 400 mm 100 mm according to section 2.1. Each sam-ple was placed between 400 mm 400 mm 11 mm oriented strand boards (OSB) to make the surfaces in contact with the hot and cold sides flat (Fig. 2). A polyethylene plastic film was also placed between the cold side and the sample to avoid condensation and alteration of the measurement values. Type-T (Cooper- Con-stantan) thermocouples were used to measure the temperature of the sample on both hot and cold sides, the OSB surface and the air (Fig. 2). In addition, a flow sensor (F-005 T-3 Concept Engi-neering) was located at the barycenter of the sample on the hot side. The flow sensor and thermocouples measurements were stored every 15 min using a battery-powered 8-channel data log-ger (OM-CP-OCTTEMP). Thermal transmittance (K) and thermal conductivity (k) were calculated based on these measurements. The data were considered once the stationary regime was estab-lished with a thermal difference between the hot side and the cold side greater than or equal to 20 °C. Equations (1) and (2) were used for the calculation.

$$K = \frac{Q}{\Delta T} \quad (1)$$

Where K is thermal transmittance (W/m^2 °C); Q is the exchanged heat (W/m^2) measured with the thermal flow sensor; and ΔT is the temperature difference between both sides of the sample (°C) measured with the internal thermocouples in contact with the sample.

$$k = K \cdot e \quad (2)$$

Where k is thermal conductivity in W/m °C; K is the thermal trans-mittance (W/m^2 °C); and e is the thickness of the sample (m).

2.3. Characterization of flammability and ignitability

Two tests were carried out to evaluate the reaction of the insu-lation material to different risk situations. The glow wire test (GWT) was used to simulate overloading and overheating of elec-trical wiring. The single-flame source test was performed to mea-sure the spread of a small flame that simulated the initial ignition stage in a fire. Samples of 150 mm 150 mm 100 mm or 90 mm 250 mm 40 mm were prepared for the GWT and the single-flame source test, respectively.

The tests evaluated the material described in section 2.1. (type 1 samples) and eight more samples supplemented with different coatings and additives as detailed in Fig. 3. For type 2 samples a **fire-retardant additive for paints (MINIT Ignifugos of Pablo Minotti** based on Na_2O and SiO_2) was added to the binder mixture. Type 3–7 samples were coated with standard synthetic enamel (composed of 66.2% alkyd resin, 23% Stoddard solvent, inert fillers and pigments) or latex (composed of 68.5% acrylic polymer, 13% water, 12% $CaCO_3$, inert fillers, pigments, additives and coalescing agents) mixed with solutes such as sand, borax ($Na_2[B_4O_5(OH)_4]$ $8H_2O$), or fire-retardant additive. Some of these materials were used as fire retardants in previous studies [45]. Finally, type 8 sam-ples were coated with latex and type 9 samples were coated with cement prepared with the components and proportions of the bin-der mixture. All of the above coatings were applied manually with a brush in a single layer, with the option of using a standard elec-tric paint spray equipment for latex application on type 8 samples.

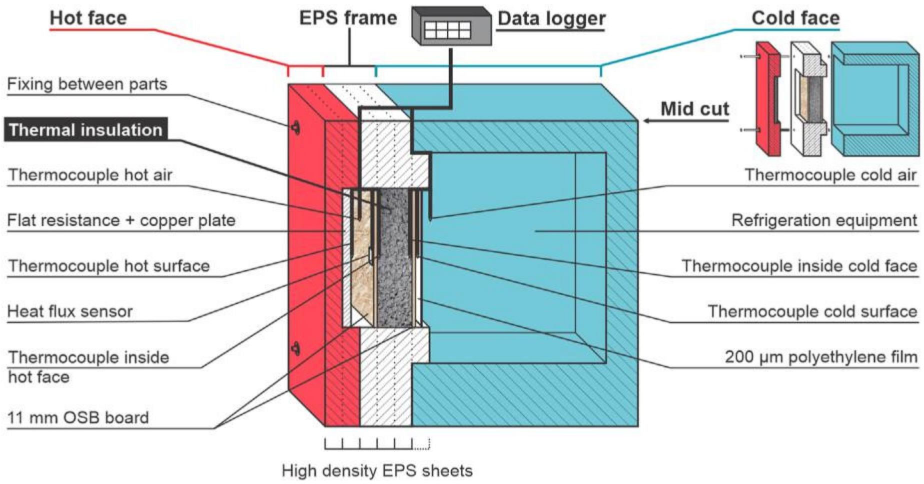

Fig. 2. Components of the hot box testing apparatus developed to measure thermal conductivity and thermal transmittance.

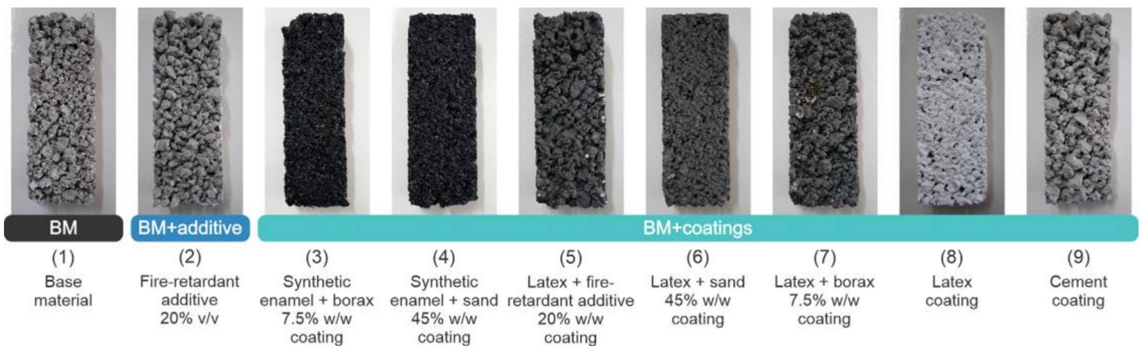

Fig. 3. Types of samples used in the single flame source test.

2.3.1. Glow wire test

Sample flammability was evaluated applying a glow wire at a discrete temperature according to the guidelines of the Interna-tional Standard IEC 60695-2-11 for final products [46]. The appara-tus used for this test (shown in Fig. 4.) was provided by a collaborating team, which developed it based on this standard.

As an initial procedure for testing, a 70 gr/cm^2 filter paper was placed under the sample. Subsequently, the glow wire was applied to the samples, initially at 840 LC and then at 960 LC for a period of 30 s in each case. In each application it was registered if ignition occurred, i.e. if the flame lasted 5 s or more. In addition, times required for ignition and flame extinction were recorded. Sample consumption by the flames and the capacity of detached particles to ignite the filter paper located below were also recorded. The test was carried out on one sample of each type. The sample was con-sidered to pass the test if the paper under the sample did not ignite and ignition did not occur or if the flames were extinguished within 30 s after the application of the glow wire.

2.3.2. Single-flame source test

Sample ignitability was evaluated according to ISO 11925-2 [47]. A test apparatus was designed according to this standard for the application of a small flame on the surface of the samples that were suspended in vertical position by a support structure (Fig. 5). The ignition source was a standard jewelry butane torch with an adjustable flame that was supplied by a gas bottle. The torch was placed on a mobile support inclined at 45L and aligned with the center of the test sample. The support was moved closer

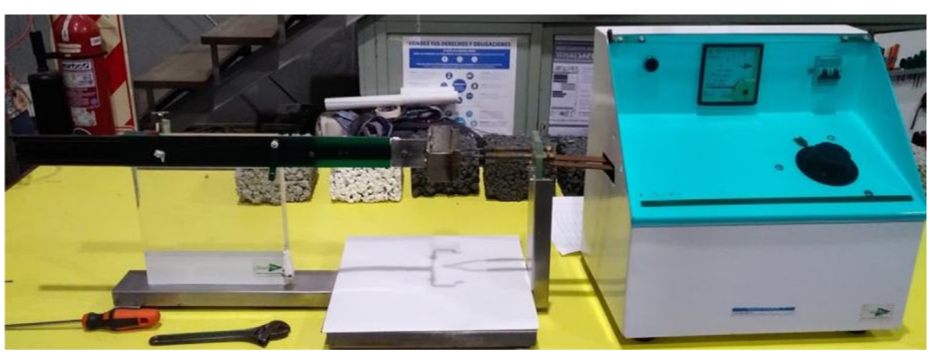

Fig. 4. GWT measuring apparatus developed according to the IEC 60695-2-11 Standard for final products [46].

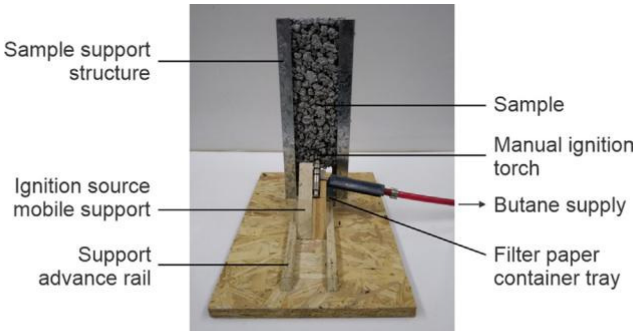

Fig. 5. Components of the single-flame source test measuring apparatus developed according to ISO 11925-2 [47].

to the sample until the torch was 4 cm above from the bottom edge of the sample and 5 mm from its surface.

Before the test, a line was drawn on each sample at a height of 190 mm from one of its edges. Two layers of 70 g/m^2 filter paper were placed in a container tray located below the sample. The torch flame was adjusted for each application on a black back-ground so that its height was 20 mm. The torch was applied to the sample for 30 s. During this period, the following data were recorded: ignition occurrences (flames lasted a period longer than 3 s); if the tip of the flame reached 150 mm above the point of application of the source, exceeding the line previously drawn and the moment this event occurred; and whether the presence of flaming droplets or detached particles caused the filter paper to ignite. The test was repeated 3 times for each type of sample (Fig. 3). The results obtained were evaluated according to UNE-EN 13501–1 [48] standard, which allowed an initial classification of the reaction to fire of the different samples.

2.4. Characterization of mechanical properties

The developed material was intended for non-load bearing applications such as wall insulation and similar applications. These applications correspond to category I of EPS defined by the ISO 4898 [49] standard. Two mechanical tests were carried out to eval-uate if the material met the minimum requirements for this cate-gory, to guarantee handling resistance. Samples of the material described in section 2.1 with a dimension of

250 mm 100 mm 20 mm and 100 mm 100 mm 50 mm were evaluated for bending and compression tests, respectively. The sample production process included sanding the pouring face to achieve the required dimensions. A thin layer of binder was applied with a brush on this face, to unify the conditions of all the faces. Samples were tested using an Instron TTBML machine provided by a collaborating team (Fig. 6).

2.4.1. Bending test

The three-point bending test was carried out in accordance with ISO 1209-1 [50] and the amendments of ISO 4898 [49]. The dis-tance between sample supports was 200 mm. The load was applied to the center of the sample with a speed of 50 mm/min until fail-ure. The deflection and bending load at the time of failure were recorded.

2.4.2. Compression test

The compression test was determined according to ISO 844 [51]. The samples were compressed between a fixed and a mobile plate that moved at a speed of 5 mm/min until each sample reached 85% of its original thickness. The force at 10% deformation was recorded to calculate the compressive stress (r_{10}).

2.5. Characterization of water absorption

To characterize sample water absorption, different samples were immersed in water for 17 h to simulate their response to extreme weather events. The immersion period was determined based on the time that water remained into the houses after the 2013 flood occurred in La Plata, region where this investigation was carried out [52].

The test was carried out on 9 specimens, one of each variant described in section 2.3. with a dimension of 150 mm 150 mm 100 mm. Firstly, the initial weight of each of the specimens was recorded before the test. Changes in color, texture and shape in the samples were qualitatively evaluated dur-ing and after immersion. Each sample was weighed and allowed to air dry after the immersion period. Sample weighs were recorded after 3 and 6 days of immersion. Finally, collected data were used to quantify the percentages of water absorption and loss in each type of sample.

2.6. Characterization of weathering behavior

This test analyzed changes that could occur in the material if it were stored or applied outside without a suitable coating. A sample with a dimension of 400 mm 400 mm 100 mm of the base material described in section 2.1 was used for the test. This sample was exposed for 10 days to external climatic conditions. The char-acteristics of the sample before and after the exposure were eval-uated both qualitatively and quantitatively. Changes in the color,

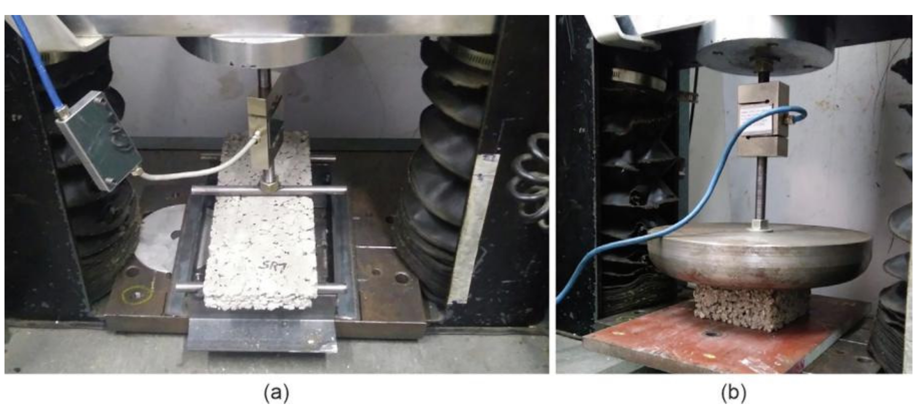

Fig. 6. Samples during bending (a) and compression tests (b).

Fig. 7. Cutting tests with manual saw (a) and a circular bench saw (b).

texture and shape of the sample were recorded. The changes in the insulating capacity of the sample after climatic exposure were also evaluated, comparing the initial and post-exposure measurement of thermal conductivity carried out once the weight of the sample was stable.

2.7. Characterization of machinability

The machinability of the material was determined by perform-ing cutting and drilling procedures that might be necessary dur-ing the application of the insulation to houses. The cuts were tested with a hand saw and a circular bench saw (Fig. 7). The per-forations were made with an electric drill carrying 3.2 mm, 4.9 mm, 6.1 mm and 7.7 mm diameter drill bits. The ease of cut-ting and the finish achieved in these procedures were evaluated according to the time and labour required for the execution, as well as the amount of material detached and the resulting neat-ness, respectively.

3. Results and discussion

3.1. Thermal properties

Thermal conductivity was measured on three base material samples manufactured according to section 2.1. Each sample was tested with a hot box apparatus based on ISO 8990 [44] standard. The thermal conductivity ranged between 0.0603 and 0.0706 W/m K. These values were calculated with equations (1) and (2) from the measurements collected shown in Table 2. The base material showed a mean bulk density of 168 ± 7 kg/m^3. The variability obtained in thermal conductivity and density could be associated with the manual methods used to manufacture the samples. There could be variations in the amount of material poured into the moulds that would modify the compression, the proportion of tight air and the insulation capacity of each sample.

The thermal conductivity and bulk density values of the base material were compared with those of conventional and alterna-tive insulating materials (Fig. 8). The thermal conductivity of the base material is comparable to that of certain conventional insulat-ing materials (vermiculite) [9] and lower than that of certain alter-native insulating materials (hemp shives, cotton stalks, date palm, oil palm, corn cob, pecan, leather scraps, sunflower, durian, typha fibers, sansevieria fiber, and banana with polypropylene fiber) [12,15,16,22]. Most of these alternative insulating materials have a higher bulk density range than the base material.

Base material performance can also be compared to studies of recycled EPS-based composites (Fig. 8). For example, Laukaitis et al. [39] obtained a compound with a conductivity coefficient between 0.060 and 0.064 W/m K, whereas Madariaga et al. [38] described plaster-based compounds for thermal insulation panels with a thermal conductivity ranging between 0.063 and 0.067 W/ m K. In addition, Sayadi et al. [34] evaluated different lightweight concrete compounds with a minimum thermal conductivity between 0.0848 and 0.0864 W/m K. Lastly, Aciu et al. [40] described compounds for fillings and thermal insulation panels with a thermal conductivity of 0.06469–0.09914 and 0.09914 W/ m K respectively. These thermal conductivity values are similar

Table 2
Measurements obtained in 3 samples tested with the hot box apparatus.

Measurements	Units	Sample 1	Sample 2	Sample 3
Hot side	LC			
Average air temperature		51.57 ± 1.08	42.89 ± 0.94	47.51 ± 0.85
Average surface temperature		48.13 ± 0.41	41.99 ± 0.99	43.68 ± 0.54
Average temperature inside the specimen		44.22 ± 0.38	39.99 ± 0.70	40.15 ± 0.54
Cold side	LC			
Average air temperature		9.13 ± 1.51	8.22 ± 1.21	12.90 ± 0.82
Average surface temperature		1.66 ± 0.37	2.99 ± 0.82	2.50 ± 0.63
Average temperature inside the specimen		2.32 ± 0.44	3.66 ± 0.71	2.57 ± 0.58
Flow	mV			
Mean of flow sensor measurements	W/m^2	2.15 ± 0.40	2.13 ± 0.35	2.26 ± 0.26
Average heat flow through the sample		25.27 ± 4.70	24.98 ± 4.16	26.55 ± 3.12
Thermal conductivity	W/m K			
Global (Insulation + OSB)		0.0507 ± 0.0094	0.0639 ± 0.0096	0.0575 ± 0.0067
Insulation		0.0603 ± 0.0113	0.0687 ± 0.0113	0.0706 ± 0.0084

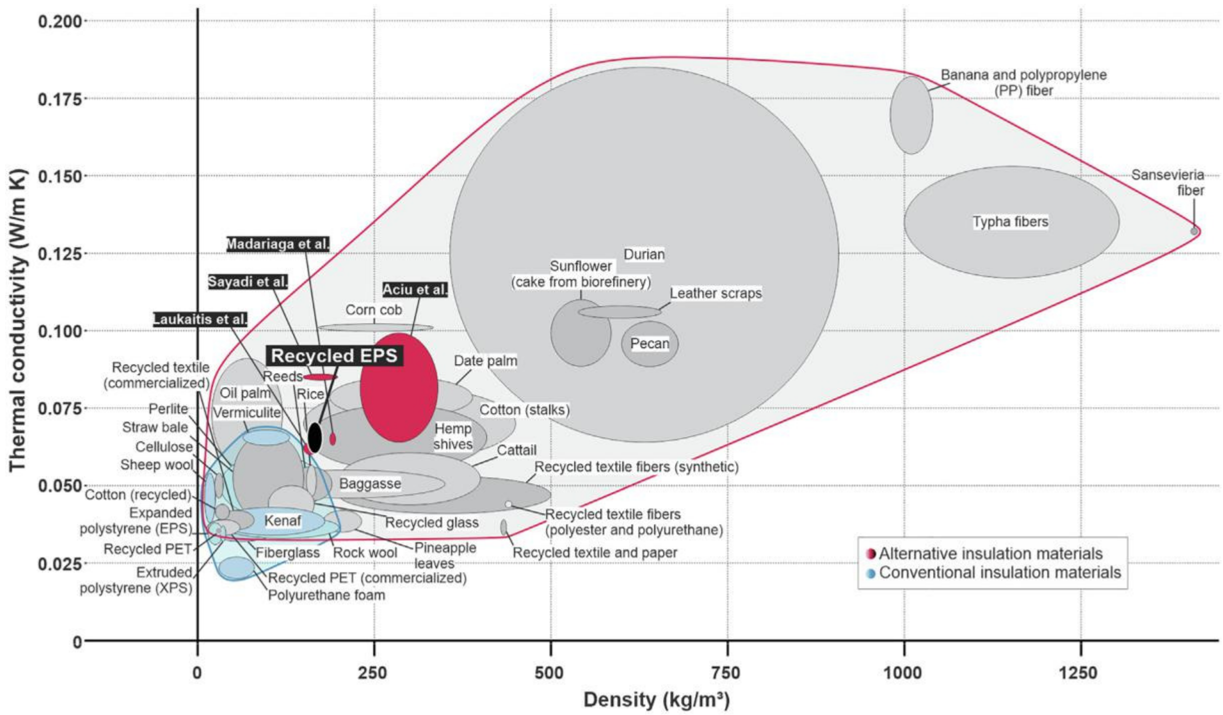

Fig. 8. Ashby's graph based on data collected from previous research [9,12,15,16,22,34,38–40]. The black ellipse indicates our results.

and higher than those obtained by the base material, suggesting this material presents appropriate thermal conductivity required for the thermal insulation uses proposed.

3.2. Flammability and ignitability

The material described in section 2.1. (type 1 samples) and eight more samples supplemented with different coatings and additives, as detailed in section 2.3., were tested for flammability and ignitability. The flammability of these materials was tested using a glow wire applied at 840 LC and 960 LC according to the International Standard IEC 60695-2-11 for final products [46]. All nine sample types evaluated passed the application of glow wire at 840 LC and 960 LC. Fig. 9 shows the response of the different samples during the application of glow wire at both temperatures. The samples passed the test at 840 LC because the wire did not ignite either the material neither the filter paper located below the sample. The base material sample (type 1 sample) and samples coated with latex and fire retardant additive (type 5 sample), latex (type 8 sample) and cement (type 9 sample) did not ignite or detach particles ignited during application of the 960 LC glow wire. The wire at 960 LC ignited the base material samples with fire retardant additive (type 2 sample) and the enamel and latex coated samples with sand and borax aggregates (types 3, 4, 6 and 7 sam-ples). Table 3 presents the response of samples that ignited. It was notable that in these samples the ignition was extinguished quickly and there were no particles that ignite the filter paper. Therefore, even these types of samples passed the test at 960 LC.

The ignitability of all sample types presented in Fig. 3 was eval-uated following the ISO 11925-2 single flame source test [47]. Table 4 presents the response of the test samples according to the evaluated parameters. Ignition occurred in the base material samples (type 1 samples) but the flame had little spread due to the cement acting as a retardant. This flame retardant effect of cement in mixtures with EPS was also reported in other investiga-tions [40]. It was recorded that as the EPS granules were consumed,

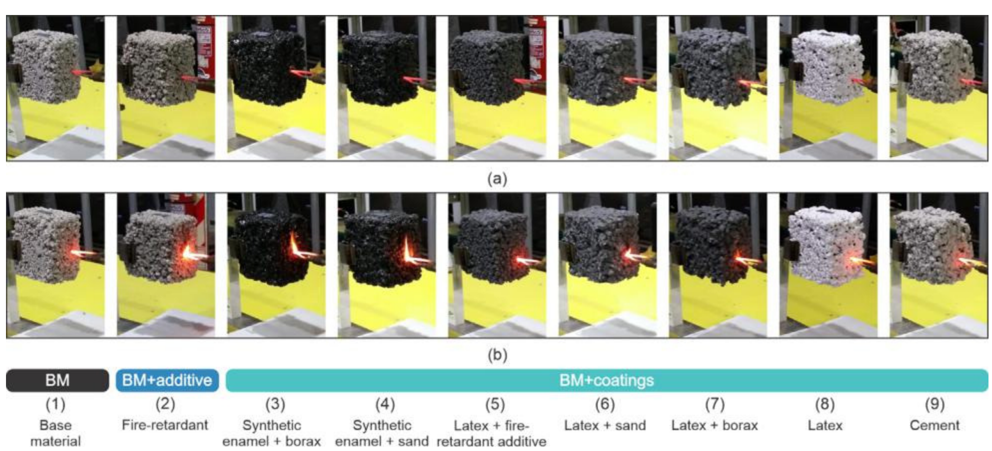

Fig. 9. Sample responses to the application of a glow wire according to the International Standard IEC 60695-2 [46]. a) Application of the glow wire at 840 LC. b) Application of the glow wire at 960 LC.

Table 3
Response to glow wire tests at 960 LC for samples in which ignition occurred: (2) Base material (BM) with fire-retardant additive; (3) BM coated with synthetic enamel and borax; (4) BM coated with synthetic enamel and sand; (6) BM coated with latex and sand; (7) BM coated with latex and borax.

Parameters	Type samples				
	(2)	(3)	(4)	(6)	(7)
Time it took for ignition to occur	2 s	4 s	<1 s	2 s	2 s
Time when the flames were extinguished	1 s	2 s	<1 s	3 s	<1 s
If the flames consumed the sample	No	No	No	No	No
If the filter paper was lit	No	No	No	No	No

Table 4
Sample responses to single flame source tests. (1) Base material (BM). (2) BM with fire-retardant additive. (3) BM coated with synthetic enamel and borax. (4) BM coated with synthetic enamel and sand. (5) BM coated with latex and fire retardant additive. (6) BM coated with latex and sand. (7) BM coated with latex and borax. (8) BM coated with latex.
(9) BM coated with cement.

Parameters	Type samples								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Does ignition occur?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Do flames reach 15 cm?	No	No	Yes	Yes	No	No	No	No	No
Time to reach 15 cm	–	–	5 s	2 s	–	–	–	–	–
Is the filter paper igniting?	No	No	No	No	No	No	No	No	No

the binder was released as non-ignited particles. In addition, the flame propagation height did not exceed 150 mm above the appli-cation point during the 30 s of application of the torch, so that the

Table 5
Fire classification of alternative and conventional insulation materials [12,22].

Material	Fire classification
Rock wool	A1-A2
Expanded Polystyrene (EPS)	E
Extruded Polystyrene (XPS)	E
Kenaf	B2
Sheep wool	B1-B2
Reeds	E
Cotton (recycled)	E
Recycled glass (commercialized)	A1
Recycled PET (commercialized)	B
Recycled textile (commercialized)	E,F
Leather scraps	E or higher classes
Recycled EPS (type 1,2,5,6,7,8,9 samples)	E or higher classes

base material could obtain at least an E classification according to the UNE-EN 13501-1 Standard [48]. Class E materials are capable of resisting for a short period, a small flame attack without sub-stantial flame spread. The final classification of the materials pre-sented in this work will depend on complementary tests, which could define a surpassing classification. As a comparison, some conventional insulation materials such as EPS and extruded poly-styrene (XPS), as well as non-conventional ones: reeds, recycled cotton, recycled textiles [12] and leather scraps [22] also present fire classification E (Table 5).

In addition to the base material, other types of samples studied could also obtain E class, except types 3 and 4 with coatings of syn-thetic enamel with borax and sand respectively, in which the flame exceeded the maximum allowed extension. Fig. 10 shows flame maximum heights and the resulting affected area that character-ized the 3 repetitions carried out for each variant. Samples coated with latex and fire retardant additive (type 5 samples); latex and sand (type 6 samples); latex (type 8 samples); and cement (type 9 samples) were the samples that had the least area affected by

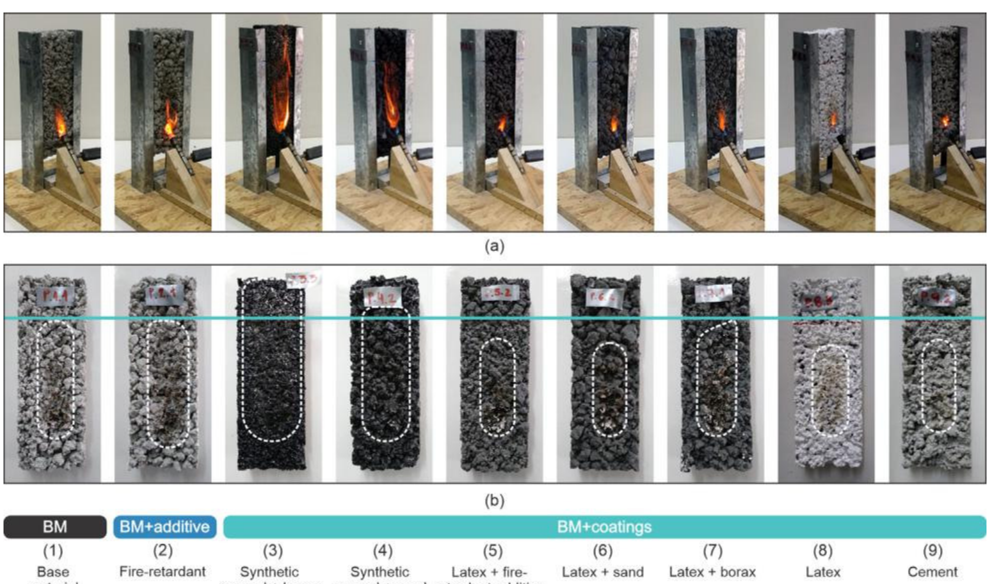

Fig. 10. Samples during and after the application of a single flame source according to ISO 11925-2 [47]. a) Maximum height of the flame reached during the 30 s of torch application. b) Affected area after exposure.

the flame and the least amount of detached particles. Therefore, these coatings improved the performance of the base material samples. Among these coatings, latex stands out, applied in type 8 samples. Latex is a coating that is easy to achieve and which application can be speeded up by using unsophisticated equip-ment. The application of this coating may be appropriate in cases where additional fire protection is required due to the conditions in which the insulating material will be applied.

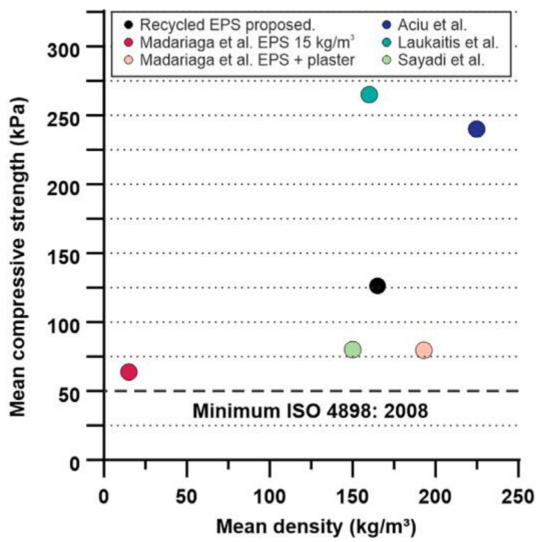

Fig. 11. Comparison of the compressive strength obtained with previous investi-gations of recycled EPS-based compounds used for thermal insulation. The black dot indicates our results.

3.3. Mechanical properties

The material described in section 2.1 was tested for bending and compression in accordance with ISO 1209-1 [50] and ISO 844 [51], respectively. The mean bending load at break of the base material was 18.3 ± 3.6 N, with a mean deflection of 3.5 ± 0.8 mm. The mean compressive strength of the base material at 10% of deformation was 126.4 ± 20.2 kPa. These results were compared with those required according to ISO 4898 [49] for category I EPS, with a minimum density of 15 kg/m^3 for non-load bearing applications. The minimum bending load at break of 15 N was slightly exceeded whereas the minimum compressive strength of 50 kPa obtained at 10% strain was more than doubled, suggesting the material shows appropriate mechanical properties for this category.

Other investigations of compounds with EPS made similar mechanical tests, providing opportunities for comparisons, although it should be considered that the methodologies followed different standards in each case. Only Madariaga et al. [38] reported the maximum bending load of 29.7 N for EPS of 15 kg/m^3 and between 7.5 N and 83.3 N for different insulating mixtures with plaster and EPS. In their research the resistance to compres-sion at 10% deformation was also evaluated for EPS and EPS with plaster binder. The compressive strength values reported in [34,38–40] are presented in Fig. 11. Our material showed a com-pressive strength inside the range of the values for previously reported materials.

3.4. Water absorption

The nine types of specimens of the base material and the vari-ants described in section 2.3 with a dimension of 150 mm 150 mm 100 mm were immersed for 17 h for the test.

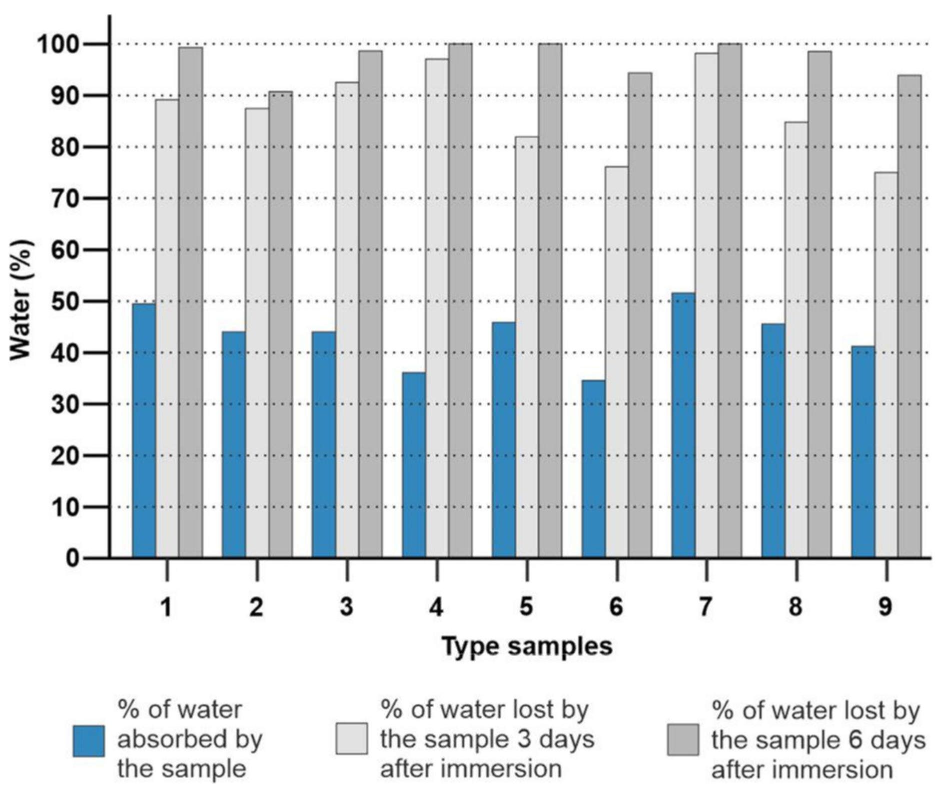

Fig. 12. Percentage of water absorption and loss after immersion for samples composed of: (1) Base material (BM); (2) BM with fire-retardant additive; (3) BM coated with synthetic enamel and borax; (4) BM coated with synthetic enamel and sand; (5) BM coated with latex and fire retardant additive; (6) BM coated with latex and sand; (7) BM coated with latex and borax. (8) BM coated with latex; (9) BM coated with cement.

The percentage of water absorption of the samples ranged from 35% to 50% after the immersion (Fig. 12). The base material sample (type 1 sample) had a water absorption percentage of 50% and a water loss rate higher than the average of all the samples tested. Test samples with additives and coatings absorbed a lower per-centage of water than the base material, except for the latex and borax coating (sample type 7) that absorbed 51% of water. All tested samples lost > 75% and > 90% of the absorbed water after 3 and 6 days of immersion respectively.

Uncoated samples changed their color during immersion although no changes in shape or structure were recorded. After immersion, the sample of the base material (type 1 sample) and the sample with fire-retardant additive (type 2 sample) lost resis-tance, tending to detach surface particles. Nevertheless, coated samples performed better without shedding particles.

As a comparison, the values obtained in this test are much higher than the 2.9% of water absorbed by the 15 kg/m^3 EPS after 96 h of immersion according to previous research [38]. The mix-ture with the cementitious binder and the porous structure of the material could be the main factors that cause it to have higher water absorption values.

3.5. Weathering behavior

A sample with a dimension of 400 mm 400 mm 100 mm of the base material described in section 2.1 was exposed for 10 days to external climatic conditions for testing. The tested sample retained its shape and strength without losing particles during or after exposure. The thermal conductivity of the sample was 0.0603 W/m K before the test and 0.0636 W/m K 10 days after the end of the period of exposure to external conditions. This result suggests that the uncoated material could have a good perfor-mance if it was stored or placed in unfavorable conditions for a short period.

3.6. Machinability

Cutting and drilling procedures were performed on the material for testing. The cutting tasks were carried out with ease and simple labour due to the lightness of the material. Surface particles were more easily detached from the material during hand saw cutting than when using a circular bench saw. With both saws, the quali-ties achieved were adequate according to the characteristics of each tool. The drilling tasks were carried out quickly without effort, with little material detachment and good resulting quality (Fig. 13). These results would indicate that the material could be easily adapted to the surfaces to insulate, using the cutting tools evaluated and it could be drilled to be fixed for house applications.

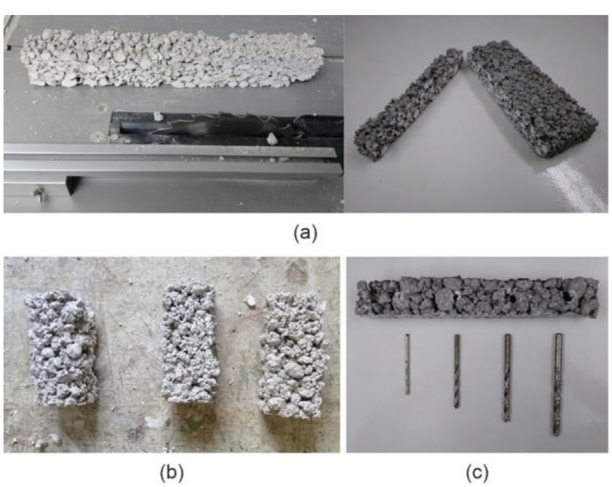

Fig. 13. Samples after cutting and drilling tasks performed. a) Samples cut with a bench saw. b) Samples cut with a manual saw. c) Sample with perforations of different diameters.

4. Conclusions

The results of the characterization tests of the base material and its variants with coatings and additives allowed us to reach the fol-lowing conclusions. The coefficient of thermal conductivity obtained (0.0603–0.0706 W/m K) positions the base material as an insulator of intermediate performance comparable to that of certain conventional and alternative insulation materials. This insulating property may remain stable if the material is stored out-doors, as long as the adverse weather conditions are not very intense or long lasting.

The flammability test showed that the base material and its variants could resist electrical overheating to 960 LC. The ignitabil-ity test showed that the base material would not present risks in the spread of a small flame, thus it could obtain at least an E clas-sification according to the UNE-EN 13501-1 standard. In cases with more demanding requirements, an improvement in the perfor-mance of the base material could be achieved with the application of certain coatings, among which latex stands out with an optimal fire response. These coatings could also help reduce the amount of water absorbed and preserve the structure of the material.

The mechanical characterization of the base material showed that the bending load at break (18.3 N) and the compressive strength obtained at 10% deformation (126.4 kPa) exceeded the minimum requirements for category I EPS. This category is suitable for non-load bearing applications. In addition, the water absorption rate obtained suggests that the material should be avoided for applications in direct contact with water.

Insulation performance and machinability of the material sug-gest the proposed material can be used feasibly for various applica-tions. Modular insulating panels are one of the applications under development for homes that already have a building envelope. Sandwich panels are other applications under study to achieve non-structural enclosures. These applications would allow the improvement of housing conditions in popular settlements through simple materials and methods. The use of heating and cooling systems would be reduced, and periods of thermal comfort would be extended while EPS waste is reused.

CRediT authorship contribution statement

Laura Elena Reynoso: Conceptualization, Investigation, Methodology, Supervision, Writing - original draft, Writing - review & editing. Ángeles Belén Carrizo Romero: Conceptualiza-tion, Investigation, Methodology. Graciela Melisa Viegas: Concep-tualization, Methodology, Funding acquisition, Supervision, Writing - original draft. Gustavo Alberto San Juan: Conceptualiza-tion, Methodology, Funding acquisition, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing finan-cial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to thank the Electrical Testing and Measurement Laboratory (LEME) of the Faculty of Engineering of the National

University of La Plata, which provided the equipment for the glow wire test (GWT) and technically assisted on experiments with this equipment, in particular to Ricardo Dias, Cándido Condorí and Juan Carlos Alfaro. We also would like to thank the Group of Applied Mechanical Tests (GEMA) of the Faculty of Engineering of the National University of La Plata, where the compression and bend-ing tests of the material were carried out, in particular to Facundo Pasquevich, Juan Ignacio Gastiarena and Felipe Robla for their tech-nical assistance in experiments. This research was funded by the National Council of Scientific and Technological Research of Argen-tina and the National University of La Plata.

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